Inference types annotation guidelines

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1 Introduction

1.1 Purpose of this annotation guide

The goal of our task is to annotate the inference mechanisms that are needed to determine if the statement entails or contradicts the premise.

1.2 The Natural Language Inference task

Natural Language Inference (NLI) is the task of determining if a statement can be inferred from a given premise. In our case, the possible outcomes are either an Entailment or a Contradiction.

1.3 Tools and materials

1.3.1 The NLI4CT dataset

The NLI4CT corpus is freely available and consists of a collection of English breast cancer Clinical Trial Reports (CTR) taken from clinicaltrials.gov. This task uses NLI for clinical trials with several use cases, such as checking that a patient complies with the trial's eligibility criteria or checking that a claim can be deduced from the trial's results. NLI4CT comprises two kinds of instances: single, where only 1 CTR is involved to perform the inference, and comparison, where 2 CTRs are needed to be compared. A premise consists of a section of a CTR (or 2 CTRs if comparison) and a statement of 1 or 2 sentences. The model should predict whether the statement entails or contradicts the premise. The task involves several kinds of inference, both involving general domain and biomedical knowledge.

1.3.2 How to run Doccano

We use Doccano to annotate the NLI4CT dataset. To use it, follow the installation instructions as well as the following:

1. If using Anaconda: create a suitable environment to install and run Doccano.
2. In a first terminal:
   (a) conda activate doccano *name of your conda environment*
   (b) doccano webserver –port 8000
3. In a second terminal:
   (a) conda activate doccano *name of your conda environment*
   (b) doccano task
4. In a web browser, open `http://127.0.0.1:8000/`.
5. Log in to the interface using the passwords defined when installing doccano (probably *admin* and *pass*).

2 How to perform the annotation

We decided to annotate only the inference mechanisms that allow solving the inference relation. In other words, we do not annotate pieces of text that do not help to solve this inference relation. Following this workflow:

1. By default, we process the statement from left to right, but at first not in detail. Instead, our first goal is to obtain the overall structure of the statement. We also check whether the task-related expressions (see Section 3.4) refer to parts that do exist in the premise. To do this, we:
   (a) Notice the task-related expressions
      - Check whether they refer to something that exists in the premise (e.g., the statement mentions Cohort 2, but there is not Cohort 2 in the premise)
   (b) Notice the connectors: conjunction, disjunction, logical contrast (whereas, however → entail conjunction)
      - Build the logical structure of the statement. Some examples:
        - A and B
          → AND(A, B)
        - a, b, c, and d are among the adverse events of the first trial
          → AND(a ∈ AE1, b ∈ AE1, c ∈ AE1, d ∈ AE1)
   (c) Notice the logical structure of the statement, with a more detailed processing of the contents of the statement
      - If the statement has the form "A and B",
        we need to prove that both A and B are true to prove Entailment
        (and as soon as either A or B is false, we stop annotating and conclude Contradiction)
      - If the statement has the form "A or B"
        we need to prove that both A and B are false to prove a Contradiction
        (and as soon as either A or B is true, we stop annotating and conclude Entailment)
   (d) Stop

3 Annotation scheme

In the following sections, we will go through the inference labels, addressing them per top-level category. These labels are summarized in Figure 1.

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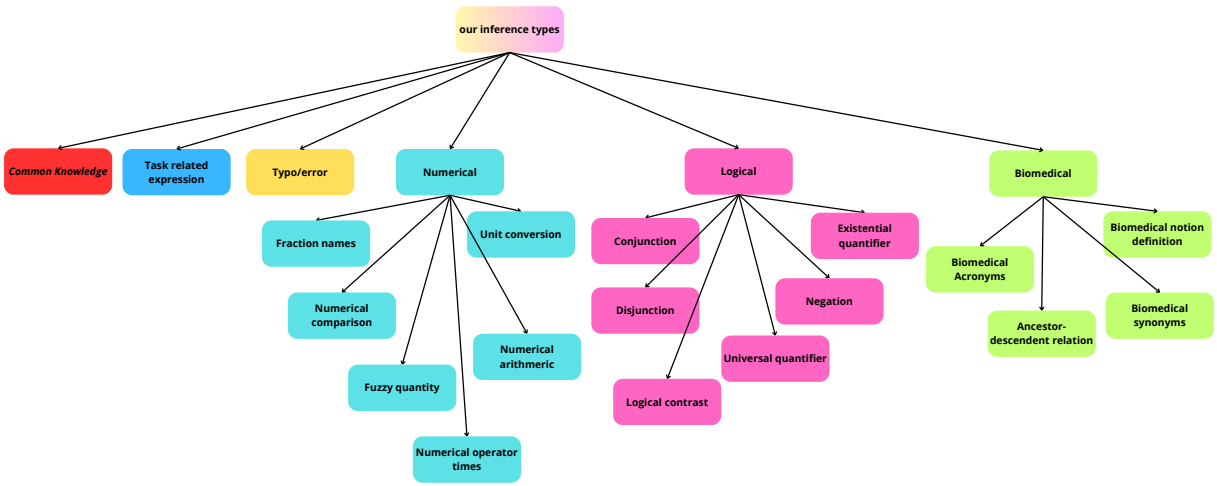

Figure 1: Inference labels' hierarchy

3.1 Logical (super-class)

Logical connectives and quantifiers, as specified below. See also `https://en.wikipedia.org/wiki/First-order_logic`.

3.1.1 Conjunction

Some terms in the statement express a logical AND.

Words to annotate: *and, both, as well as, additionally*
See also below the logical Contrast, e.g., *however*.

3.1.2 Disjunction

Some terms in the statement express a logical OR.

Words to annotate: *or*

3.1.3 Negation

A term in the statement asserts that some assertion is not true.

Words to annotate: *no, not, none, neither*
Special case for *none* → existential quantifier + negation.
Special case for *neither* → conjunction + negation.

3.1.4 Contrast

Some terms in the statement express a contrast between two clauses.

Words to annotate: *whereas, however, but, on the other hand, the difference*
In all cases, a logical contrast between two clauses also expresses a conjunction of these two clauses. Therefore, a Conjunction label must also be added. (recall that with a conjunction between two or more clauses, we have to examine the truthfulness of each of the clauses to prove Truth)

3.1.5 Existential quantifier

A term in the statement asserts that a condition applies to at least one element in a specified set (∃ symbol).

Words to annotate: *there exists, there are/is, at least one, some...*

3.1.6 Universal quantifier

A term in the statement asserts that a condition applies to all the elements in a specified set ($\forall$ symbol).

Words to annotate: *all, any, every*

3.2 Numerical (super-class)

Fine-grained numerical inference types that include arithmetic operations, conversions, comparisons, and manipulation of numbers and quantities.

3.2.1 Numerical comparison

When two numbers or quantities need to be compared and check that the number mentioned in the statement.

Words to annotate: *more than, less than, at least*

3.2.2 Numerical arithmetic

When an arithmetic operation (addition, subtraction, multiplication, division) is needed to solve the inference relation.

Words to annotate: Statement: *The dosage of Denosumab [...] is 12 times higher than the dosage of Tamoxifen.* Premise - primary trial: *Denosumab 120 mg Q4W* Premise - secondary trial: *10 mg once daily from 5th day to 25th day of menstrual cycle for 3 months*

3.2.3 Unit conversion

A conversion from one unit to another unit is required to solve the inference relation.

Words to annotate: *months* vs. *weeks* vs. *days.*
stage 4 $\equiv$ *stage IV*
Special case for units of time (*days, months, years, etc.*) where the label Common knowledge can also be used.

3.2.4 Fuzzy quantity

A quantity subject to several interpretations is present in the statement, and its interpretation is needed to solve the inference relation.

Words to annotate: *most commonly, several*

3.3 Biomedical (super-class)

Biomedical terms and knowledge that are needed to solve the inference relation.

3.3.1 Ancestor-descendant relation

Terms that are either children or parents of the considered term in the premise.

Words to annotate: *Gastro-intestinal adverse event* (statement) and *Vomiting* (premise). *drugs* (statement) and *prochlorperazine*, *granisetron hydrochloride* (premise).

3.3.2 Biomedical acronym

Acronyms that shorten a biomedical term or expression. This acronym should be listed in one of the UMLS terminologies.

Words to annotate: *IV* (Intravenous), *AE* (Adverse Event), *BMI* (Body Mass Index).

3.3.3 Biomedical notion definition

Instances that require knowing the definition of a medical term and what properties it implies. Therefore, we need to compare these properties with the information in the premise.

Words to annotate: For instance: *morbidly obese* (statement) $\Rightarrow$ meaning to have a BMI greater than 40 ; *Have a BMI of 25 kg/m^2 or greater* (premise). Or: *triple negative* (premise) meaning *estrogen negative, progesterone negative and HER2 negative* (statement).

3.3.4 Biomedical synonym

Two biomedical terms, one in the statement and one in the premise that are equivalent.

Words to annotate: For instance: *Paraplatin* is the brand name of *Carboplatin*. Carboplatin was present in the premise while Paraplatin was used in the statement.

3.4 Task-related expression

Any expression that refers to clinical trials or the structure of a clinical trial report.

Words to annotate: *primary/secondary trial, patient, adverse event, cohort, intervention, eligible, inclusion/exclusion criteria*

3.5 Typo/error

We annotate a typo or error in the statement or the premise that would lead to a major misunderstanding of the aforementioned and have an impact on the statement-premise relation resolution.

Words to annotate: For instance: "$110,034\approx$í$\neg\int L$" $\rightarrow$ here the typo makes the measurement unit unreadable.

3.6 Generic knowledge

"Generic knowledge" (opposed to domain-specific knowledge) are notions that do not fall into the previous categories. In our case, it is often temporal words (*months*, *hours*) or concepts such as age, gender, ethnicity, nationality, etc.

Words to annotate: For instance: In the statement "*There are no conditions on gender...*". And in the premise "*Inclusion Criteria: Women 18 years...*".

4 Training examples

Example 1: Fig. 2:

- **Task related expression**: *primary trial, eligible* and *patients*
- **Biomedical acronym**: *BMI*
- **Biomedical notion definition**: we need to know the BMI value for morbidly obese patients
- **Numerical comparison**: Once we know the BMI value for morbidly obese patients, we need to compare it with the minimum BMI provided in the premise

Example 2: Fig. 3:

- **Task related expression**: *adverse event, primary trial* and *patient cohort*
- **Negation**: *None*
- **Numerical comparison**: *more than*
- **Fraction name**: *quarter*

Statement: Morbidly obese patients are eligible for the primary trial.

Premise: Inclusion Criteria:
Women and men ages 18 or older
Have been previously diagnosed with a malignant solid tumor, completed their required surgical, and/or chemotherapy and/or radiation curative intent therapy at least three months prior to enrollment, and have an anticipated treatment-free life span of 12 months or longer. Chemoprophylaxis with tamoxifen or aromatase inhibitors for breast cancer in women and anti- Luteinizing hormone-releasing hormone (LHRH) therapy for prostate cancer in men will be permitted.
Have a BMI of 25 kg/m^2 or greater and weight <=400 lbs.

Figure 2: Example n°1

Statement: None of the individual adverse event types recorded in the primary trial affected more than a quarter of the patient cohort.

Premise: Adverse Events 1:
Total: 5/14 (35.71%)
Ileus 1/14 (7.14%)
General disorders and administration site conditions - Other, disease progression 2/14 (14.29%)
Infections and infestations - Other, pneumonia 1/14 (7.14%)
Acute kidney injury 1/14 (7.14%)

Figure 3: Example n°2

Example 3: Fig. 4:

- **Task related expression**: *intervention, primary/secondary trial*
- **Contrast**: *whereas*
- **Ancestor-descendant relation**: we need to know that *ZD1839* is a drug
- **Conjunction**: *and*
- **Existential quantifier**: *consists of, requires at least 3* (equivalent to "there is/are")
- **Biomedical acronym**: *Sm-153 = Samarium-153*

Statement: The intervention in the primary trial consists of a single drug, ZD1839, whereas in the secondary trial the intervention requires at least 3 different drugs, including Zoledronic acid, Samarium-153 and pegfilgrastim.

Premise: Primary premise:
INTERVENTION 1:
Gefitinib (ZD1839)
ZD1839 at a daily dose of 250 mg or 500 mg depending on final dose in parent trial

Secondary premise:
INTERVENTION 1:
Zoledronic Acid
Zoledronic acid, vitamin D and calcium supplements.
INTERVENTION 2:
Zoledronic Acid + Radiopharmaceuticals
Zoledronic acid, vitamin D and calcium supplements, plus Sr-89 or Sm-153.

Figure 4: Example n°3

Example 4: Fig. 5:

- **Task related expression**: *patient, primary trial, excluded*
- **Disjunction**: *or*

- **Unit conversion**: *years* and *weeks*
- **Numerical comparison**: *2 years* and *4 weeks*
- **Universal quantifier**: *any*

Statement: Any patients that have undergone Antineoplastic chemotherapy, Androgens, Estrogens or Progestogens therapy in the last 2 years before study entry are excluded from the primary trial.

Premise:
Exclusion Criteria:
Any of the following current (=< 4 weeks prior) or planned therapies:
Antineoplastic chemotherapy (anti-HER2 agents allowed)
Androgens
Estrogens (any delivery route)
Progestogens

Figure 5: Example n°4

Example 5:

- **Task related expression**: *patient, primary trial*
- **Numerical arithmetic**: compute the difference between *Ki67 expression at baseline: 15* and *Ki67 expression at 6 weeks: 10* $\Rightarrow$ 5%.
- **Numerical comparison**: comparing the *5%* in the statement with the result of the computation done for *Numerical arithmetic*.

Statement: Over 6 weeks of TAK-228 Plus Tamoxifen treatment patients in the primary trial experienced a 5% reduction in the Percentage of cells with Ki67 expression

Premise:
[...]
Overall Number of Participants Analyzed: 23
Median (Inter-Quartile Range)
Unit of Measure: Percentage of cells with Ki67 expression Ki67 expression at baseline: 15 (10 to 25)
Ki67 expression at 6 weeks: 10 (2 to 38)

Figure 6: Example n°5

Example 6:

- **Task related expression**: *primary trial candidates*
- **Ancestor-descendant relation**: *eating disorders* and *anorexia*
- **Negation**: *not*
- **Fuzzy quantity**: *common*
- **Numerical comparison**: comparing the *1/50 (2%)* with *not common*

Statement: Eating disorders were not common for the primary trial candidates

Premise:
Adverse Events 1:
Total: 17/50 (34.00%)
[...]
Anorexia 1/50 (2.00%)
[...]

Figure 7: Example n°6